MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3510 COUNTER CURRENT DECANTATION THICKENER FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management company(PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub-supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1. INTENT

This document is to Prepare for design of **Counter Current Decantation Thickener (3510)** structure's foundation. The analysis & design is carried out in accordance with project specifications,codes & standards. The design calculation ensures the safety and serviceability of the framing system.

2. SCOPE

In the specified area, there are 3 nos of thickener i.e. THK-001,THK-002 & THK-003.As per load data, the loads are same for all 3 thickeners and the height of pedestals are also at same level from FGL. Therefore design calculation has been done considering the given loads at top of pedestal. The analysis and design has been carried in excel, considering the load data attached here with in Attachment 1. The load combinations for serviceability check and member's strength calculation are as per SBC.

3. DESIGN REQUIREMENTS

3.1 Reference Documents & Codes

Reference Documents:

i) Doc. No. MD-216-1000-EG-ST-CRT-1001 [Civil,Structural and Architectural Design Criteria]
ii) Doc. No. MD-101-SMEM-EG-ST-CRT-1001 [Structural Design Criteria - Process Building & Structures]
iii) Doc. No. MD-101-SMEM-EG-CV-CRT-1001 [Civil Design Criteria - General]

Reference Codes:

i) SBC 301 - 2007
ii) SBC 304-CR-18
iii) ACI 318
iv) ACI SP-17M(09)

3.2 Material Property

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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3.3 Grouting

- Grout shall be cementitious under steel column bases of Strength 40 MPa

3.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

3.5 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to Sea Water or Soil	Foundation, Column & Tie Beam	75
Exposed to weather	Column & Floor Beam	50
	Slab	50

4. DESIGN CRITERIA

4.1 Design Loads

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3510-EG-ME-GAD-1002/1004/1006) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

4.2 Load Combinations

Load combinations are based on SBC - 301.
Basic Load Cases are –

1. Tank Dead Load & Bridge/Drive Dead Load(D)
2. Max. Fluid Load (F_{max})
3. Normal Operational Fluid Load ($F_{operating}$)
4. Seismic Load (E)
5. Wind Load (W)
6. Bridge/Drive Live Load(L)

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Allowable Stress Design

Load Case	Load Combination
101	$D+F_{max}$
102	$D+F_{max}+L$
103	$D+F_{max}+0.75L$
104	$D+F_{max}+W$
105	$D+F_{operating}+0.7E$
106	D+F+0.75L+0.75W
107	$D+F_{operation}+0.75L+0.75*(0.7E)$
108	0.6D+W
109	0.6D+0.7E

Ultimate Strength Design

Load Case	Load Combination
201	$1.4(D+F_{max})$
202	$1.2(D+F_{max})+1.6L$
203	1.2D+f1L
204	1.2D+0.8W
205	1.2D+1.6W +f1L
206	1.2D+1.0E +f1L
207	0.9D+1.6W
208	0.9D+1.0E

4.3 Stability against Overturning & Sliding

Overturning

Structures and buildings foundations, shall have the following minimum stability ratio against overturning:

- 1.5 For load combinations with wind or seismic loads.
- 2.0 For all other load combinations.

Sliding

A minimum factor of safety against sliding shall be 1.5.

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4.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 460 kN/m^2 at 1.2m depth below EGL (RL 911.30m) as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A01, bore hole BH-12.

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5. DESIGN CALCULATION OF REINFORCEMENT

5.1 Inner Column

5.1.1 Load Calculation :

Input load data for Counter Current Decantation CCD Thickener

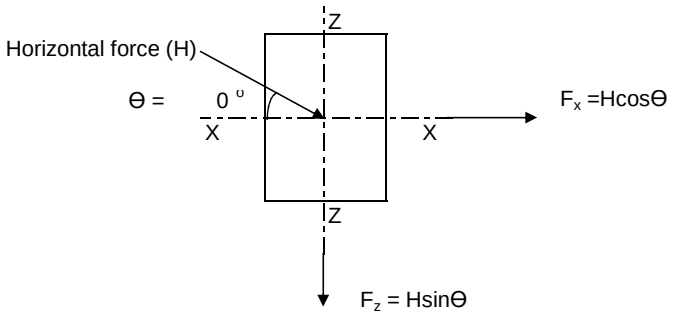
Refer OutoTec drawing number : 900707_GAUO1_ST61_MLD15_35101/35102/35103 rev 1

Load Table :

	Tank dead load	Max fluid load		Normal operating fluid load		Seismic load			Wind load			Bridge/drive dead load	Bridge/drive live load
Central Column	kN	kN		kN		kN	kN	kN-m	kN	kN	kN-m	kN	kN
	V	H	V	H	V	H	V	Moment	H	V	Moment	V	V
	40	0	2410	0	2017	10	0	9	7	0	7	0	0

H : Horizontal load V : Vertical load

Horizontal reactions are in the plane of bracing.



Depth of foundation from U/S of base plate = 1.6 m

Load at bottom of foundation level

Primary load cases		F_x	F_y	F_z	M_x	M_z
		kN	kN	kN	kN-m	kN-m
Tank dead load	D	0	40	0	0	0
Selfweight of column = 0.6*0.6*(1.6-0.55)*25 =		0	9.45	0	0	0
Selfweight of footing =		0	85.9375	0	0	0
Weight of soil above footing =		0	69	0	0	0
Bridge/drive dead load		0	0	0	0	0
Max fluid load	F_{max}	0	2410	0	0	0
Normal operating fluid load	$F_{operati}$	0	2017	0	0	0
Seismic load	E	10	0	0	0	25
Wind load	W	7	0	0	0	18.2
Bridge/drive live load	L	0	0	0	0	0

Combinations including seismic load to use normal operating fluid load.

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Load combinations for foundation servicibility check :

Sl no	Basic combinations as per cl 2.4.1 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z
		kN	kN	kN	kN-m	kN-m	kN-m
101	$D+F_{max}$	0	2614	0	0	0	0
102	$D+F_{max}+L$	0	2614	0	0	0	0
103	$D+F_{max}+0.75L$	0	2614	0	0	0	0
104	$D+F_{max}+W$	7	2614	0	0	0	18.2
105	$D+F_{operating}+0.7E$	7	2221	0	0	0	17.5
106	D+F+0.75L+0.75W	5.25	2614	0	0	0	13.7
107	$D+F_{operation}+0.75L+0.75*(0.7E)$	5.25	2221	0	0	0	13.1
108	0.6D+W	7	150	0	0	0	18.2
109	0.6D+0.7E	7	150	0	0	0	17.5

Weight of soil above footing includes in all load combination except (0.6D+W or 0.6D+E).

Load combinations for strength check :

Sl no	Basic combinations: as per O19045-001-C-K-24400-001-02 cl 4.2.9	F_x	F_y	F_z	M_x	M_y	M_z	q_{net1}	q_{net2}	q_{net3}	q_{net4}	q_{avg}
		kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2	kN/m^2	kN/m^2	kN/m^2
201	$1.4(D+F_{max})$	0	3443	0	0	0	0	550.9	551	551	551	551
202	$1.4(D+F_{max})+1.7L$	0	3443	0	0	0	0	550.9	551	551	551	551
203	1.2D+f1L	0	59	0	0	0	0	9.494	9.49	9.49	9.49	9.49
204	1.2D+0.8W	5.6	59	0	0	0	14.56	15.09	3.9	15.1	3.9	15.1
205	1.2D+1.6W +f1L	11.2	59	0	0	0	29.12	20.68	-1.69	20.7	-1.69	20.7
206	1.2D+1.0E +f1L	10	59	0	0	0	25	19.09	-0.11	19.1	-0.11	19.1
207	0.9D+1.6W	11.2	44.51	0	0	0	29.12	18.3	-4.06	18.3	-4.06	18.3
208	0.9D+1.0E	10	44.51	0	0	0	25	16.72	-2.48	16.7	-2.48	16.7

f_1 = 1 cl 2.4.1 SBC 301

Design Parameter for net base pressure calculation:

Area of Footing , A = 6.25 m^2

Zz-z = 2.6 m^3

Zx-x = 2.6 m^3

Net Pressure, $q_{net} = P/A \pm M_z/Z_{zz} \pm M_x/Z_{xx}$

Consider average net pressure for design.

q_{avg} for design = 550.9 kN/m^2

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5.1.2 Design of Column C1 :

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 75 mm

d' = 95 mm

Y_h = (h-d')/h = 0.68

Y_b = (b-d')/b = 0.68

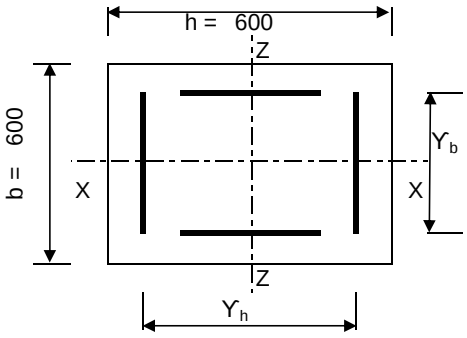
Unsupported length of column parallel to X-X axis, l_{ux} = 1.3 m kl_{ux}/b = 4.333 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.3 m kl_{uz}/h = 4.333

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 9746 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	kN	kN-m		kN	kN-m				kN-m			kN-m				
201	3443.2	0.0	0.0	5297	0.0	0.0	0.42	0.120	907.2	0.00	0.120	907.2	0.00	1.61	0	safe
202	3443.2	0.0	0.0	5297	0.0	0.0	0.42	0.110	831.6	0.00	0.110	831.6	0.00		0	safe
203	59.3	0.0	0.0	91.3	0.0	0.0	0.007	0.050	378	0.00	0.050	378	0.00		0	safe
204	59.3	0.0	14.6	91.3	0.0	22.4	0.007	0.050	378	0.00	0.050	378	0.06		0.01	safe
205	59.3	0.0	29.1	91.3	0.0	44.8	0.007	0.050	378	0.00	0.050	378	0.12		0.03	safe
206	59.3	0.0	25.0	91.3	0.0	38.5	0.007	0.050	378	0.00	0.050	378	0.10		0.03	safe
207	44.5	0.0	29.1	68.5	0.0	44.8	0.005	0.050	378	0.00	0.050	378	0.12		0.03	safe
208	44.5	0.0	25.0	68.5	0.0	38.5	0.005	0.050	378	0.00	0.050	378	0.10		0.03	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{lie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	12	20	+	0	0	3772	1.05	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

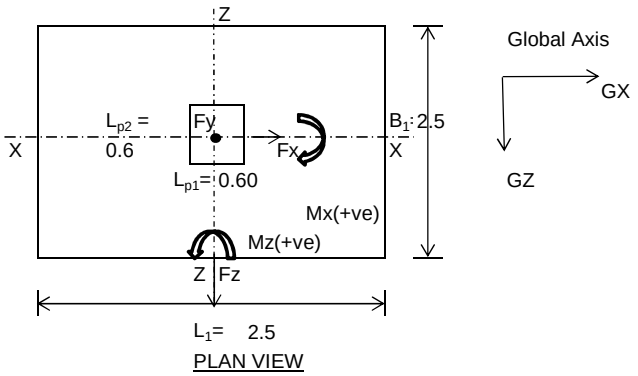
M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

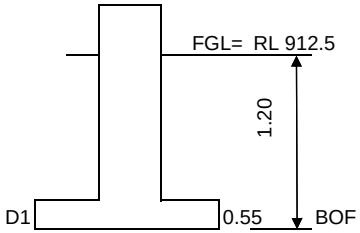
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5.1.3 Design of Foundation F1



PLAN VIEW



SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	912.5	m
EL of Highest Water Table ,HWT =	RL	909.5	m
Depth of foundation from FGL. ,H1 =	(-)	1.20	m
Ht of W.T. from FGL level =	(-)	3.00	m
Net Safe Bearing Capacity considered =		450	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	MPa
Reinforcement Grade =	f_y	420	MPa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2.5	m
Footing dimension along Z direction,B_1	=	2.5	m
Depth, D_1	=	0.55	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.6 x 0.6	=	0.36	m^2
Area of Footing , A = 2.50 x 2.50	=	6.25	m^2
Zz-z = 2.5 x 2.50 x 2.50 / 6	=	2.6	m^3
Zx-x = 2.5 x 2.5 x 2.5 / 6	=	2.6	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	85.9	kN
Wt of soil above ftg =	68.9	kN
Total weight,W=	155	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.6	m
Pedestal dimension along Z direction,Lp2=	0.6	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 1.2 x 18 = 472 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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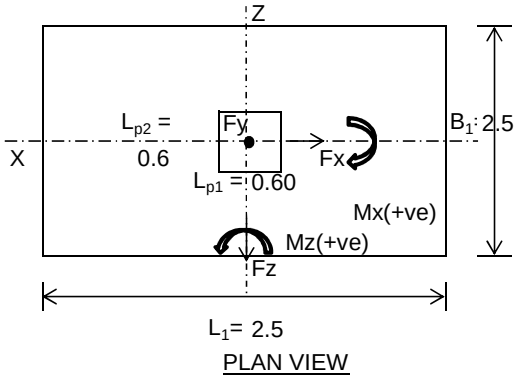
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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1	Force from Load Combinations						Bearing Pressure Calculation						Check for Overturning:								Check for Sliding					
	Axial Force			Moment									About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	NSBC increased by	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m²	FOS	kN/m²		kN/m²		kN/m²	kN/m²			kN-m	kN-m			kN	kN	kN	kN		
101	0	-2614	0	0	0	0	418.3	1	471.6	safe	418.3	ok	0.0	3268	-	-	0.0	3268	-	ok	0.0	0.0	0.0	2614	-	ok
102	0	-2614	0	0	0	0	418.3	1	471.6	safe	418.3	ok	0.0	3268	-	-	0.0	3268	-	ok	0.0	0.0	0.0	2614	-	ok
103	0	-2614	0	0	0	0	418.3	1	471.6	safe	418.3	ok	0.0	3268	-	-	0.0	3268	-	ok	0.0	0.0	0.0	2614	-	ok
104	7	-2614	0	0	0	18.2	425.3	1	471.6	safe	411.3	ok	18.2	3268	180	ok	0.0	3268	-	ok	7.0	0.0	7.0	2614	-	ok
105	7	-2221	0	0	0	17.5	362.2	1	471.6	safe	348.7	ok	17.5	2777	159	ok	0.0	2777	-	ok	7.0	0.0	7.0	2221	-	ok
106	5.25	-2614	0	0	0	13.7	423.6	1	471.6	safe	413.1	ok	13.7	3268	239	ok	0.0	3268	-	ok	5.3	0.0	5.3	2614	-	ok
107	5.25	-2221	0	0	0	13.1	360.5	1	471.6	safe	350.4	ok	13.1	2777	212	ok	0.0	2777	-	ok	5.3	0.0	5.3	2221	-	ok
108	7	-150	0	0	0	18.2	31.1	1	471.6	safe	17.1	ok	18.2	188	10.3	ok	0.0	188	-	ok	7.0	0.0	7.0	150.1	-	ok
109	7	-150	0	0	0	17.5	30.8	1	471.6	safe	17.4	ok	17.5	188	10.7	ok	0.0	188	-	ok	7.0	0.0	7.0	150.1	-	ok

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PLAN VIEW

Reinforcement Summary : in mm

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	20	217.3	200	20	207.6	200

Footing Dimension :

Footing dimension along X direction,L_1	=	2.5	m
Footing dimension along Z direction,B_1	=	2.5	m
Depth, D_1	=	0.55	m
Clear cover	=	75	mm

Column Dimension :

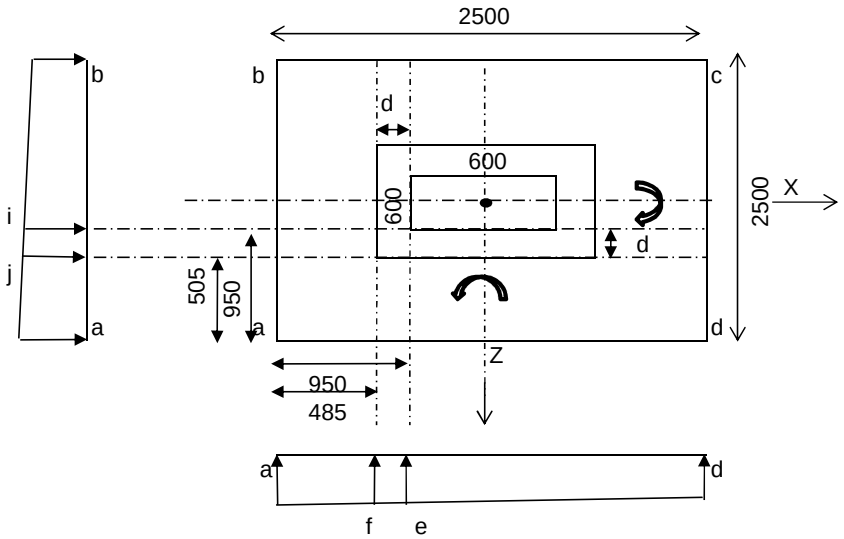
Pedestal dimension along X direction,Lp1	=	0.6	m
Pedestal dimension along Z direction,Lp2	=	0.6	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	MPa
Reinforcement Grade =	f_y	420	MPa



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Critical Load Combination for Design : 201

Maximum Net Base Pressure : 551.0 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*551]*2.5*0.95*0.95	=	621.6	kN-m
Shear force at Face "f"	= 551*0.485*2.5	=	668.1	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*551]*2.5*0.95*0.95	=	621.6	kN-m
Shear force at Face "j"	= 551*0.505*2.5	=	695.6	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2500 mm)

Cantilever Length		=	950	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	465	mm
Bending Moment at "e"		M_u =	621.6	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0056	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(465-0.0056As/2)As=621.6*10^6	A_s =	3615.1	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*2500*550	$A_{s,min}$ =	2475	mm^2
Spacing required	2500*22/28*20^2/3615.1	S_{reqd} =	217.3	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2500*22/28*20^2/200	$A_{s,pro}$ =	3928.6	mm^2
a = $A_s f_y$/0.85f'$_c$b	3928.6*420/(0.85*35*2500)	a =	22.19	mm
c = a/β_1	22.19/0.8	c =	27.74	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/27.74*(465-27.74)	ε_t =	0.047	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	668.1	kN
Moment at "f"	= [0.5*551]*2.5*0.485*0.485	M_u =	162.0	kN-m
ρ_w provided	3928.6/2500/465	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=465*2500/10^6	bd =	1.163	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	1228.5	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	1167.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1994.5	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	4279.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 2500 mm)

Cantilever Length			=	950	mm
Dia of Bar		d_b	=	20	mm
Effective Depth		d	=	445	mm
Bending Moment at "i"		M_u	=	621.6	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a	=	0.0056	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(445-0.0056As/2)As=621.6*10^6	A_s	=	3785.5	mm^2
P min			=	0.18	%
$A_{s,min}$	0.18/100*2500*550	$A_{s,min}$	=	2475	mm^2
Spacing required	2500*22/28*20^2/3785.5	S_{reqd}	=	207.56	mm c/c
Spacing Provided		S_{prov}	=	200	mm c/c
A_s Provided	2500*22/28*20^2/200	$A_{s,pro}$	=	3928.6	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	3928.6*420/(0.85*35*2500)	a	=	22.19	mm
c = a/β_1	22.19/0.8	c	=	27.74	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/27.74*(445-27.74)	ε_t	=	0.045	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u	=	695.6	kN
Moment at "j"	= [0.5*551]*2.5*0.505*0.505	M_u	=	175.7	kN-m
ρ_w provided	= 3928.6/2500/445	$\rho_{w,pro}$	=	0.0035	%
Effective Shear Area	=445*2500/10^6	bd	=	1.113	m^2
V_c = lesser of					
a) (0.16λ$\sqrt{f_c}$' + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$	=	1170.8	kN
b) (0.16λ$\sqrt{f_c}$' + 17ρ_w)bd		$V_{c,b}$	=	1119.8	kN
c) 0.29λ$\sqrt{f_c}$'bd		$V_{c,c}$	=	1908.7	kN
$\varphi(V_c + 0.66\sqrt{f_c}'bd)$			=	4097.8	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c}'bd)$ ok				ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$	=	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure			=	551.00	kN/m^2
Shear force,V_u	= 551*[2.5*2.5-(0.6+0.465)*(0.6+0.445)]		=	2830.53	kN
Shear peremeter,b_0	= 2*[(0.6+0.465)+(0.6+0.445)]		=	4.22	m
V_c =0.17(1+2/β)$\sqrt{f_c}'b_0d$	= 0.51 $\sqrt{f_c}'b_0d$				
V_c = 0.083(α_sd/b_0 + 2)$\sqrt{f_c}'b_0d$	= 0.44 $\sqrt{f_c}'b_0d$	For Edge column	α_s	30	
V_c = 0.33$\sqrt{f_c}'b_0d$	= 0.33 $\sqrt{f_c}'b_0d$				
φV_c	= 0.75*0.33*SQRT(35)*4220*465/1000		=	2873.3	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe				

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5.2 Outer Column

5.2.1 Load Calculation :

Input load data for Counter Current Decantation CCD Thickener

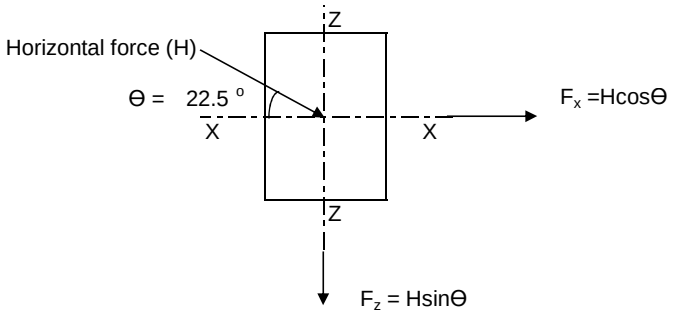
Refer OutoTec drawing number : 900707_GAUO1_ST61_MLD15_35101/35102/35103 rev 1

Load Table :

	Tank dead load	Max fluid load		Normal operating fluid load		Seismic load			Wind load			Bridge/drive dead load	Bridge/drive live load
Outer Column	kN	kN		kN		kN	kN	kN-m	kN	kN	kN-m	kN	kN
	V	H	V	H	V	H	V	Moment	H	V	Moment	V	V
	13	42	270	35	226	6	9	0	5	11	0	25	68

H : Horizontal load V : Vertical load

Horizontal reactions are in the plane of bracing.



Depth of foundation from U/S of base plate = 1.6 m

Load at bottom of foundation level

Primary load cases		F_x	F_y	F_z	M_x	M_z
		kN	kN	kN	kN-m	kN-m
Tank dead load	D	0	13	0	0	0
Selfweight of column = 0.4*0.4*(1.6-0.3)*25 =		0	5.2	0	0	0
Selfweight of footing =		0	19.2	0	0	0
Weight of soil above footing =		0	39	0	0	0
Bridge/drive dead load		0	25	0	0	0
Max fluid load	F_{max}	38.81	270	16.08	25.728	62.096
Normal operating fluid load	$F_{operati}$	32.34	226	13.4	21.44	51.744
Seismic load	E	5.55	9	2.3	3.68	8.88
Wind load	W	4.62	11	1.92	3.072	7.392
Bridge/drive live load	L	0	68	0	0	0

Combinations including seismic load to use normal operating fluid load.

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Load combinations for foundation servicibility check :

Sl no	Basic combinations as per cl 2.4.1 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z
		kN	kN	kN	kN-m	kN-m	kN-m
101	$D+F_{max}$	38.81	371	16.08	25.73	0	62.1
102	$D+F_{max}+L$	38.81	439	16.08	25.73	0	62.1
103	$D+F_{max}+0.75L$	38.81	422	16.08	25.73	0	62.1
104	$D+F_{max}+W$ (+ve)	43.43	382	18	28.8	0	69.5
105	$D+F_{operating}+0.7E$ (+ve)	36.23	334	15.01	24.02	0	58
106	D+F+0.75L+0.75W (+ve)	42.28	431	17.52	28.03	0	67.6
107	$D+F_{operation}+0.75L+0.75*(0.7E)$ (+ve)	35.25	383	14.61	23.37	0	56.4
108	0.6D+W (+ve)	4.62	87.3	1.92	3.072	0	7.39
109	0.6D+0.7E (+ve)	3.885	83	1.61	2.576	0	6.22

Weight of soil above footing includes in all load combination except (0.6D+W or 0.6D+E).

Load combinations for foundation strength check :

Sl no	Basic combinations: as per O19045-001-C-K-24400-001-02 cl 4.2.9	F_x	F_y	F_z	M_x	M_y	M_z	q_{net1}	q_{net2}	q_{net3}	q_{net4}	q_{avg}
		kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2	kN/m^2	kN/m^2	kN/m^2
201	$1.4(D+F_{max})$	54.3	438	22.5	36.02	0	86.93	351.4	96.7	246	-8.83	299
202	$1.4(D+F_{max})+1.7L$	54.3	554	22.5	36.02	0	86.93	396.5	142	291	36.3	344
203	1.2D+f1L	0	120	0	0	0	0	46.81	46.8	46.8	46.8	46.8
204	1.2D+0.8W (+ve)	3.7	61	1.54	2.458	0	5.914	35.95	18.6	28.8	11.4	32.4
205	1.2D+1.6W (+ve) +f1L	7.39	137	3.07	4.915	0	11.83	78.21	43.6	63.8	29.2	71
206	1.2D+1.0E (+ve) +f1L	5.55	129	2.3	3.68	0	8.88	68.73	42.7	57.9	31.9	63.3
207	0.9D+1.6W (+ve)	7.39	56.48	3.07	4.915	0	11.83	46.59	11.9	32.2	-2.46	39.4
208	0.9D+1.0E (+ve)	5.55	47.88	2.3	3.68	0	8.88	37.1	11.1	26.3	0.3	31.7

f_1 = 1 cl 2.4.1 SBC 301

Design Parameter for net base pressure calculation:

Area of Footing , A = 2.56 m^2

Zz-z = 0.68 m^3

Zx-x = 0.68 m^3

Net Pressure, $q_{net} = P/A \pm M_z/Z_{zz} \pm M_x/Z_{xx}$

Consider average net pressure for design.

q_{avg} for design = 343.8 kN/m^2

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5.2.2 Design of Column C2 :

Materials :

Concrete Grade, f_c' = 35 MPa

Reinforcement Grade, f_y = 420 MPa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.2 % i.e. ρ_g = 0.012

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 400 mm

Depth parallel to Z-Z axis, b = 400 mm

Clear Cover = 75 mm

d' = 95 mm

Y_h = (h-d')/h = 0.53

Y_b = (b-d')/b = 0.53

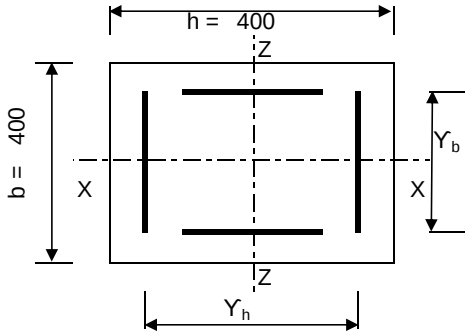
Unsupported length of column parallel to X-X axis, l_{ux} = 1.3 m kl_{ux}/b = 6.5 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.3 m kl_{uz}/h = 6.5

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 4452 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c bh$	$R_{nx} = M_{nx1}/f_c' b^2 h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c' h^2 b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	kN	kN-m		kN	kN-m				kN-m			kN-m				
201	438.5	36.0	86.9	674.6	55.4	133.7	0.12	0.075	168	0.33	0.075	168	0.80	1.61	0.86	safe
202	554.1	36.0	86.9	852.4	55.4	133.7	0.152	0.075	168	0.33	0.075	168	0.80		0.86	safe
203	119.8	0.0	0.0	184.4	0.0	0.0	0.033	0.070	156.8	0.00	0.070	156.8	0.00		0	safe
204	60.6	2.5	5.9	93.3	3.8	9.1	0.017	0.070	156.8	0.02	0.070	156.8	0.06		0.01	safe
205	137.4	4.9	11.8	211.4	7.6	18.2	0.038	0.070	156.8	0.05	0.070	156.8	0.12		0.04	safe
206	128.8	3.7	8.9	198.2	5.7	13.7	0.035	0.070	156.8	0.04	0.070	156.8	0.09		0.03	safe
207	56.5	4.9	11.8	86.9	7.6	18.2	0.016	0.070	156.8	0.05	0.070	156.8	0.12		0.04	safe
208	47.9	3.7	8.9	73.7	5.7	13.7	0.013	0.070	156.8	0.04	0.070	156.8	0.09		0.03	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars, db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
1920	4	20	+	4	16	2062	1.29	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

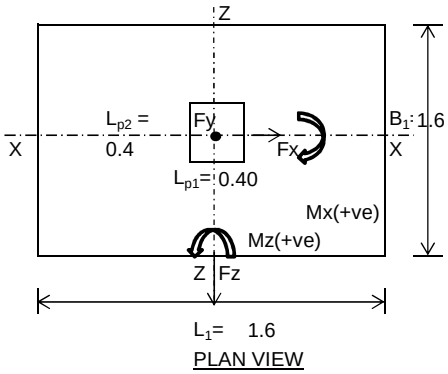
M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.6

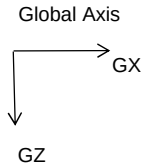
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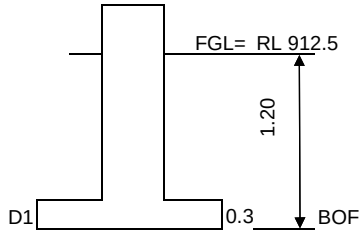
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5.2.3 Design of Foundation F2



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	912.5	m
EL of Highest Water Table ,HWT =	RL	909.5	m
Depth of foundation from FGL. ,H1 =	(-)	1.20	m
Ht of W.T. from FGL level =	(-)	3.00	m
Net Safe Bearing Capacity considered =		450	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	MPa
Reinforcement Grade =	f_y	420	MPa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	1.6	m
Footing dimension along Z direction,B_1	=	1.6	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.4 x 0.4	=	0.16	m^2
Area of Footing , A = 1.60 x 1.60	=	2.56	m^2
Zz-z = 1.6 x 1.60 x 1.60 / 6	=	0.68	m^3
Zx-x = 1.6 x 1.6 x 1.6 / 6	=	0.68	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	19.2	kN
Wt of soil above ftg =	38.9	kN
Total weight,W=	58	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.4	m
Pedestal dimension along Z direction,Lp2=	0.4	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 1.2 x 18 = 472 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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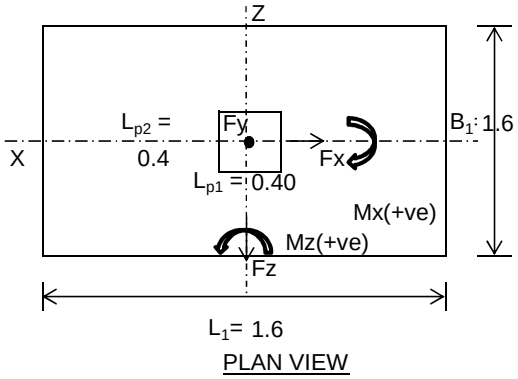
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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F2	Force from Load Combinations						Bearing Pressure Calculation						Check for Overturning:								Check for Sliding					
	Axial Force			Moment									About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	NSBC increased by	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	%	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
101	38.81	-371	16.1	25.7	0	62.1	273.7	1	471.6	safe	16.4	ok	62.1	297	4.79	ok	25.7	297	11.6	ok	38.8	16.1	42.0	371.3	9.24	ok
102	38.81	-439	16.1	25.7	0	62.1	300.3	1	471.6	safe	43.0	ok	62.1	351	5.66	ok	25.7	351	13.7	ok	38.8	16.1	42.0	439.3	10.9	ok
103	38.81	-422	16.1	25.7	0	62.1	293.7	1	471.6	safe	36.4	ok	62.1	338	5.45	ok	25.7	338	13.1	ok	38.8	16.1	42.0	422.3	10.5	ok
104	43.43	-382	18	28.8	0	69.5	293.4	1	471.6	safe	5.4	ok	69.5	306	4.41	ok	28.8	306	10.6	ok	43.4	18.0	47.0	382.3	8.5	ok
105	36.23	-334	15	24	0	58	250.4	1	471.6	safe	10.3	ok	58.0	267	4.61	ok	24.0	267	11.1	ok	36.2	15.0	39.2	333.6	8.89	ok
106	42.28	-431	17.5	28	0	67.6	308.4	1	471.6	safe	28.1	ok	67.6	344	5.1	ok	28.0	344	12.3	ok	42.3	17.5	45.8	430.5	9.83	ok
107	35.25	-383	14.6	23.4	0	56.4	266.5	1	471.6	safe	32.8	ok	56.4	306	5.44	ok	23.4	306	13.1	ok	35.3	14.6	38.2	383.0	10.5	ok
108	4.62	-87	1.92	3.07	0	7.39	49.5	1	471.6	safe	18.8	ok	7.4	70	9.46	ok	3.1	69.9	22.7	ok	4.6	1.9	5.0	87.3	18.2	ok
109	3.885	-83	1.61	2.58	0	6.22	45.2	1	471.6	safe	19.4	ok	6.2	66	10.6	ok	2.6	66.1	25.7	ok	3.9	1.6	4.2	82.6	20.5	ok

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PLAN VIEW

Reinforcement Summary : in mm

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	16	260.0	200	16	239.7	200

Footing Dimension :

Footing dimension along X direction, L_1	=	1.6	m
Footing dimension along Z direction, B_1	=	1.6	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

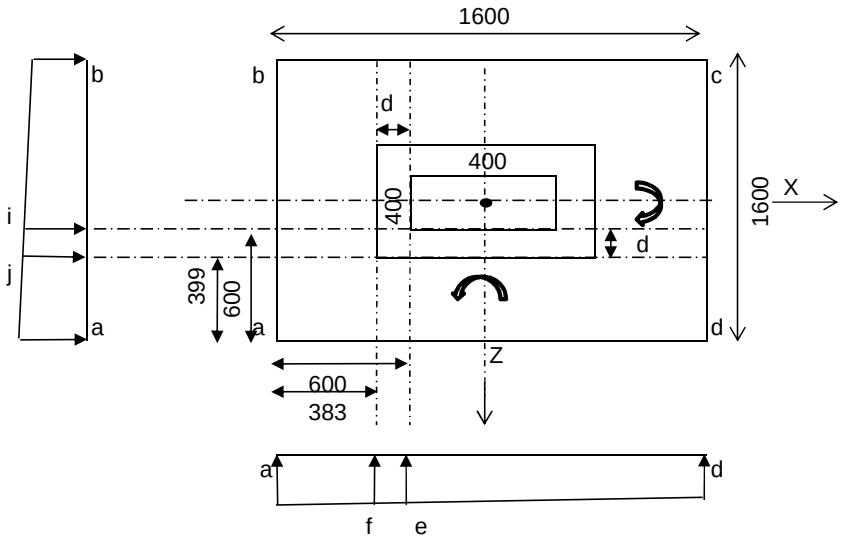
Pedestal dimension along X direction,Lp1	=	0.4	m
Pedestal dimension along Z direction,Lp2	=	0.4	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	MPa
Reinforcement Grade =	f_y	420	MPa



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Critical Load Combination for Design : 201

Maximum Net Base Pressure : 343.8 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*343.8]*1.6*0.6*0.6	=	99.0	kN-m
Shear force at Face "f"	= 343.8*0.383*1.6	=	210.7	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*343.8]*1.6*0.6*0.6	=	99.0	kN-m
Shear force at Face "j"	= 343.8*0.399*1.6	=	219.5	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 1600 mm)

Cantilever Length		=	600	mm
Dia of Bar		d_b =	16	mm
Effective Depth		d =	217	mm
Bending Moment at "e"		M_u =	99.0	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0088	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(217-0.0088As/2)As=99*10^6	A_s =	1238.0	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1600*300	$A_{s,min}$ =	864	mm^2
Spacing required	1600*22/28*16^2/1238	S_{reqd} =	260.0	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	1600*22/28*16^2/200	$A_{s,pro}$ =	1609.2	mm^2
a = $A_s f_y$/0.85f'$_c$b	1609.2*420/(0.85*35*1600)	a =	14.20	mm
c = a/β_1	14.2/0.8	c =	17.75	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/17.75*(217-17.75)	ε_t =	0.034	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	210.7	kN
Moment at "f"	= [0.5*343.8]*1.6*0.383*0.383	M_u =	40.3	kN-m
ρ_w provided	1609.2/1600/217	$\rho_{w,pro}$ =	0.0046	%
Effective Shear Area	=217*1600/10^6	bd =	0.347	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	359.6	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	356.0	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	595.7	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1283.8	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1600 mm)

Cantilever Length		=	600	mm
Dia of Bar	d_b	=	16	mm
Effective Depth	d	=	201	mm
Bending Moment at "i"	M_u	=	99.0	kN-m
$C = 0.85f_c'ba$ and $T = A_sf_y$ Therefore 'a' = $f_y/0.85f_c'b * A_s$	a	=	0.0088	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$ 0.9*420*(201-0.0088As/2)As=99.02*10^6	A_s	=	1342.7	mm^2
P min		=	0.18	%
$A_{s,min}$ 0.18/100*1600*300	$A_{s,min}$	=	864	mm^2
Spacing required 1600*22/28*16^2/1342.7	S_{reqd}	=	239.69	mm c/c
Spacing Provided	S_{prov}	=	200	mm c/c
A_s Provided 1600*22/28*16^2/200	$A_{s,pro}$	=	1609.2	mm^2
$a = A_{st}f_y/0.85f'_cb$ 1609.2*420/(0.85*35*1600)	a	=	14.20	mm
$c = a/\beta_1$ 14.2/0.8	c	=	17.75	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c) 0.003/17.75*(201-17.75)	ε_t	=	0.031	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"	V_u	=	219.5	kN
Moment at "j" = [0.5*343.8]*1.6*0.399*0.399	M_u	=	43.8	kN-m
ρ_w provided = 1609.2/1600/201	$\rho_{w,pro}$	=	0.0050	%
Effective Shear Area =201*1600/10^6	bd	=	0.322	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$	$V_{c,a}$	=	332.0	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$	$V_{c,b}$	=	331.8	kN
c) $0.29\lambda\sqrt{f_c'}bd$	$V_{c,c}$	=	551.8	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1190.6	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required	$A_{v,min}$	=	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	343.80	kN/m^2
Shear force, V_u = 343.8*[1.6*1.6-(0.4+0.217)*(0.4+0.201)]		=	752.65	kN
Shear peremeter, b_0 = 2*[(0.4+0.217)+(0.4+0.201)]		=	2.44	m
$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$				
$V_c = 0.083(\alpha_sd/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.39 $\sqrt{f_c'}b_0d$	For Edge column	α_s	30	
$V_c = 0.33\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$				
φV_c = 0.75*0.33*SQRT(35)*2436*217/1000		=	774.0	kN
Remarks : if $\varphi V_c > V_u$ ok = safe				

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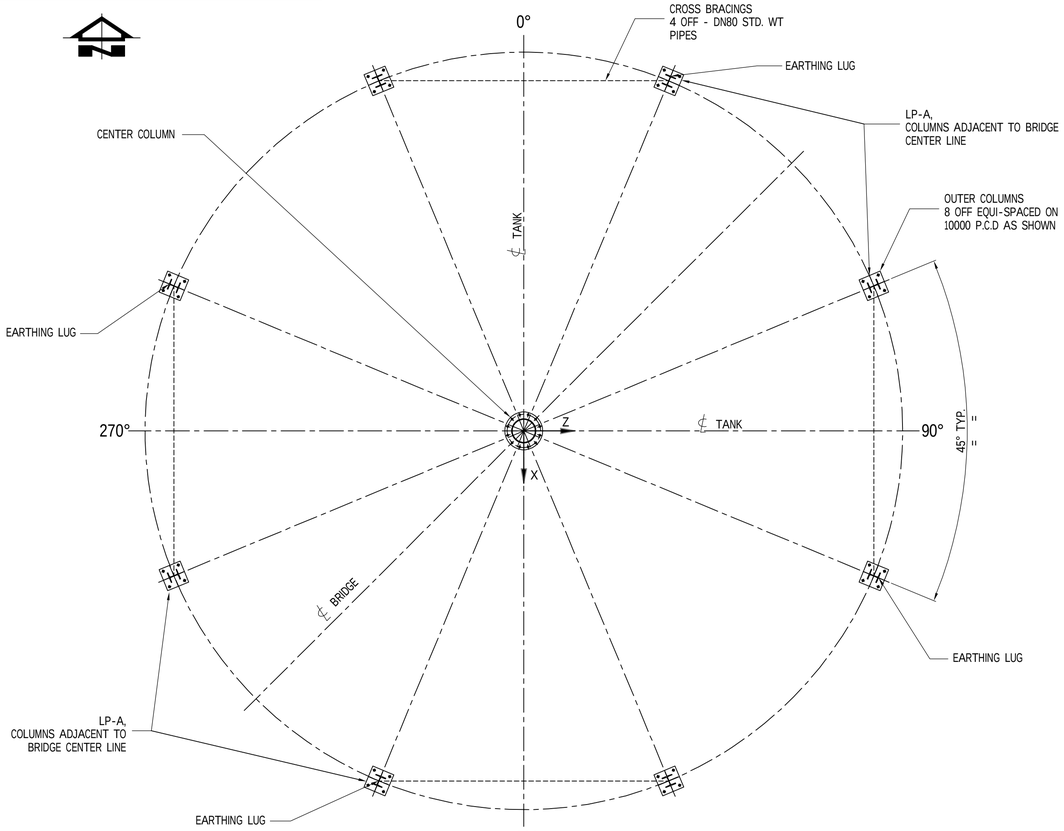
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ATTACHMENT 1 OTE INPUT DRAWINGS(FOOTING AND LOADING PLAN)

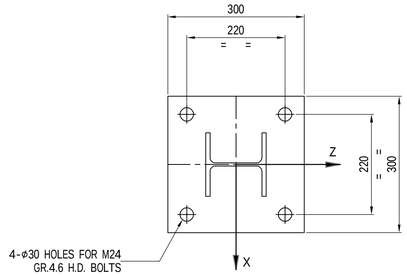
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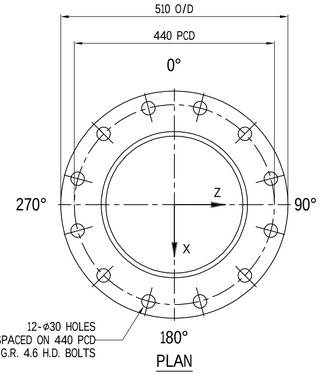
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REV: A01



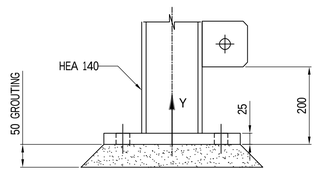
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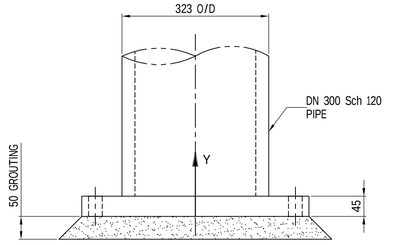
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
CENTRE COLUMN
SCALE 1:6

LOAD TABLE

		Tank Dead Load	Max Fluid Load SG 1.528		Normal Operating Fluid Load SG 1.279		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
	Column PCD (mm)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	10 000	13	42	270	35	226	6	9	0	5	11	0	25	68
Central Column	0	40	0	2 410	0	2 017	10	0	9	7	0	7	0	0
Total Applied Load	-	144	0	4 570	0	3 825	37	0	127	25	0	99	73	170

NOTES:
1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 8 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING

SEISMIC DATA TO SBC301

Ss, S1:	0.10,0.02	
Importance Factor:	1	
Site Class:	B	

WIND DATA TO SBC301

V:	43	m/s
Exposure Class:	C	

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3510-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3510-EG-ME-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02871-001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ ST61 _ MLD15 _ 35101 — DOCUMENT ID: OU601239394 — REVISION: 1

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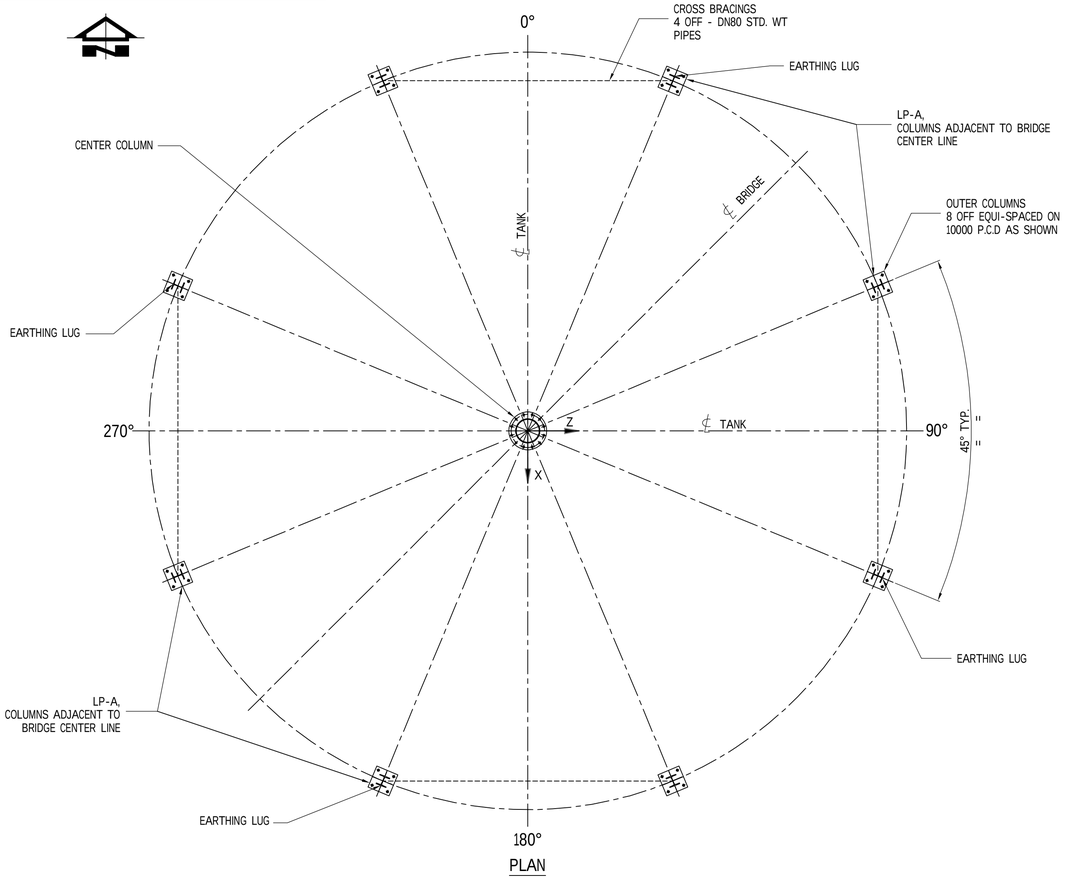
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DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

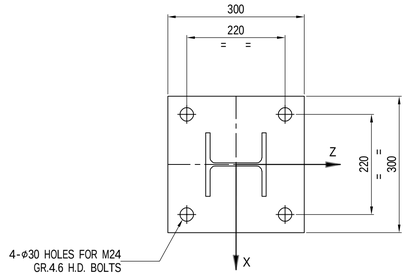
MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOADING PLAN
COUNTER CURRENT DECANTATION
3510-THK-001 T50-CCD THICKENER 1, 12m

DRAWING NUMBER: MD - 216 - 3510 - EG - ME - GAD - 1002 — REV: A01

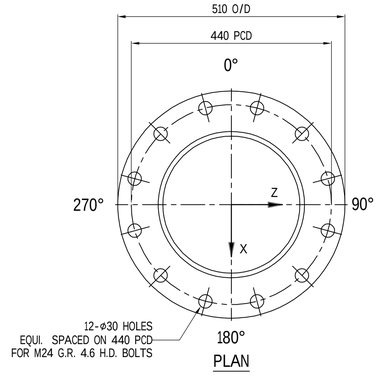
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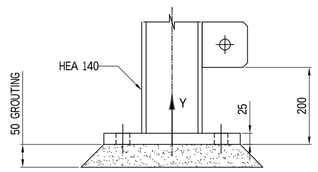
PLAN



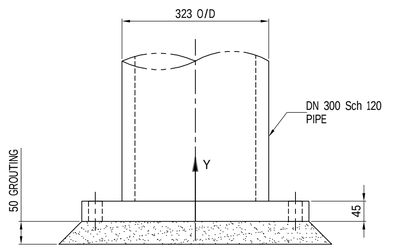
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
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Outer Columns	10 000	13	42	270	35	226	6	9	0	5	11	0	25	68
Central Column	0	40	0	2 410	0	2 017	10	0	9	7	0	7	0	0
Total Applied Load	-	144	0	4 570	0	3 825	37	0	127	25	0	99	73	170

NOTES:

1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 8 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING

SEISMIC DATA TO SBC301

Ss, S1:	0.10,0.02
Importance Factor:	1
Site Class:	B

WIND DATA TO SBC301

V:	43	m/s
Exposure Class:	C	

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3510-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3510-EG-ME-GAD-1003	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

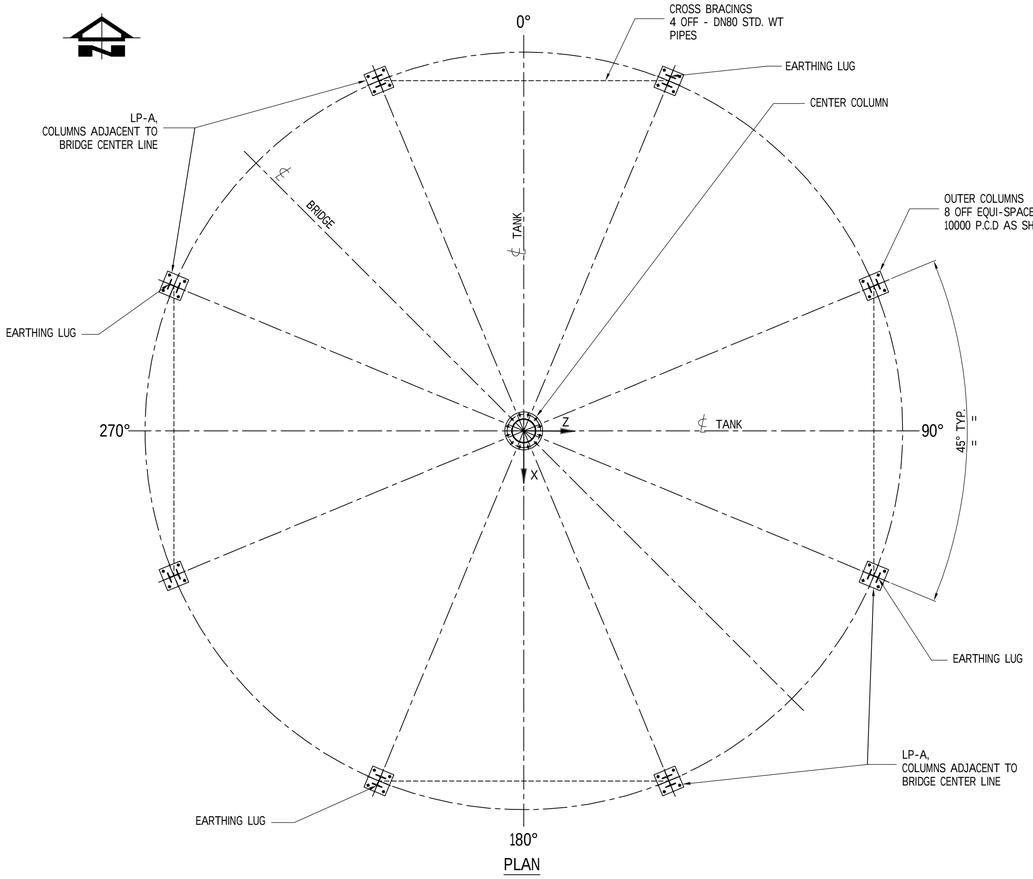
LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02871-002 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ST61 _MLD15 _35102 — DOCUMENT ID: OU601247967 — REVISION: 1

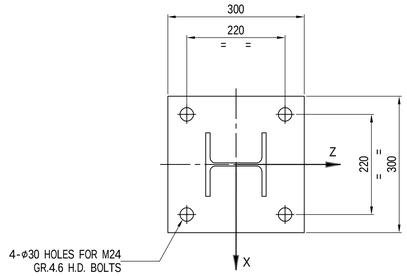
MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOADING PLAN
COUNTER CURRENT DECANTATION
3510-THK-002 T60-CCD THICKENER 2, 12m

DRAWING NUMBER: MD - 216 - 3510 - EG - ME - GAD - 1004 — REV: A01

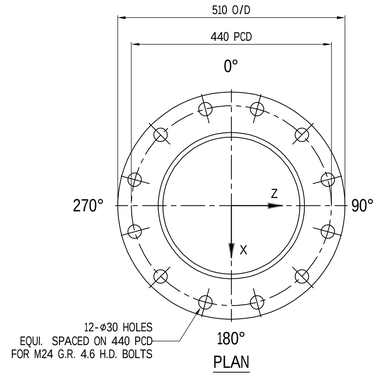
DRAWING NUMBER: MD - 216 - 3510 - EG - ME - GAD - 1006
REV: A01



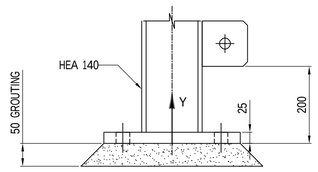
PLAN



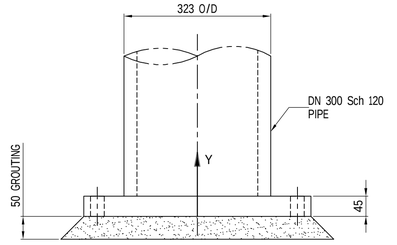
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
CENTRE COLUMN
SCALE 1:6

LOAD TABLE

		Tank Dead Load	Max Fluid Load SG 1.528		Normal Operating Fluid Load SG 1.279		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
	Column PCD (mm)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	10 000	13	42	270	35	226	6	9	0	5	11	0	25	68
Central Column	0	40	0	2 410	0	2 017	10	0	9	7	0	7	0	0
Total Applied Load	-	144	0	4 570	0	3 825	37	0	127	25	0	99	73	170

NOTES:
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10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING

SEISMIC DATA TO SBC301

Ss, S1:	0.10,0.02	
Importance Factor:	1	
Site Class:	B	

WIND DATA TO SBC301

V:	43	m/s
Exposure Class:	C	

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3510-EG-ME-DAT-1003	EQUIPMENT DATASHEET								
MD-216-3510-EG-ME-GAD-1005	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ST61 _MLD15 _35103 — DOCUMENT ID: OU601250418 — REVISION: 1

DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOADING PLAN
COUNTER CURRENT DECANTATION
3510-THK-003 T70-CCD THICKENER 3, 12m

DRAWING NUMBER: MD - 216 - 3510 - EG - ME - GAD - 1006 — REV: A01

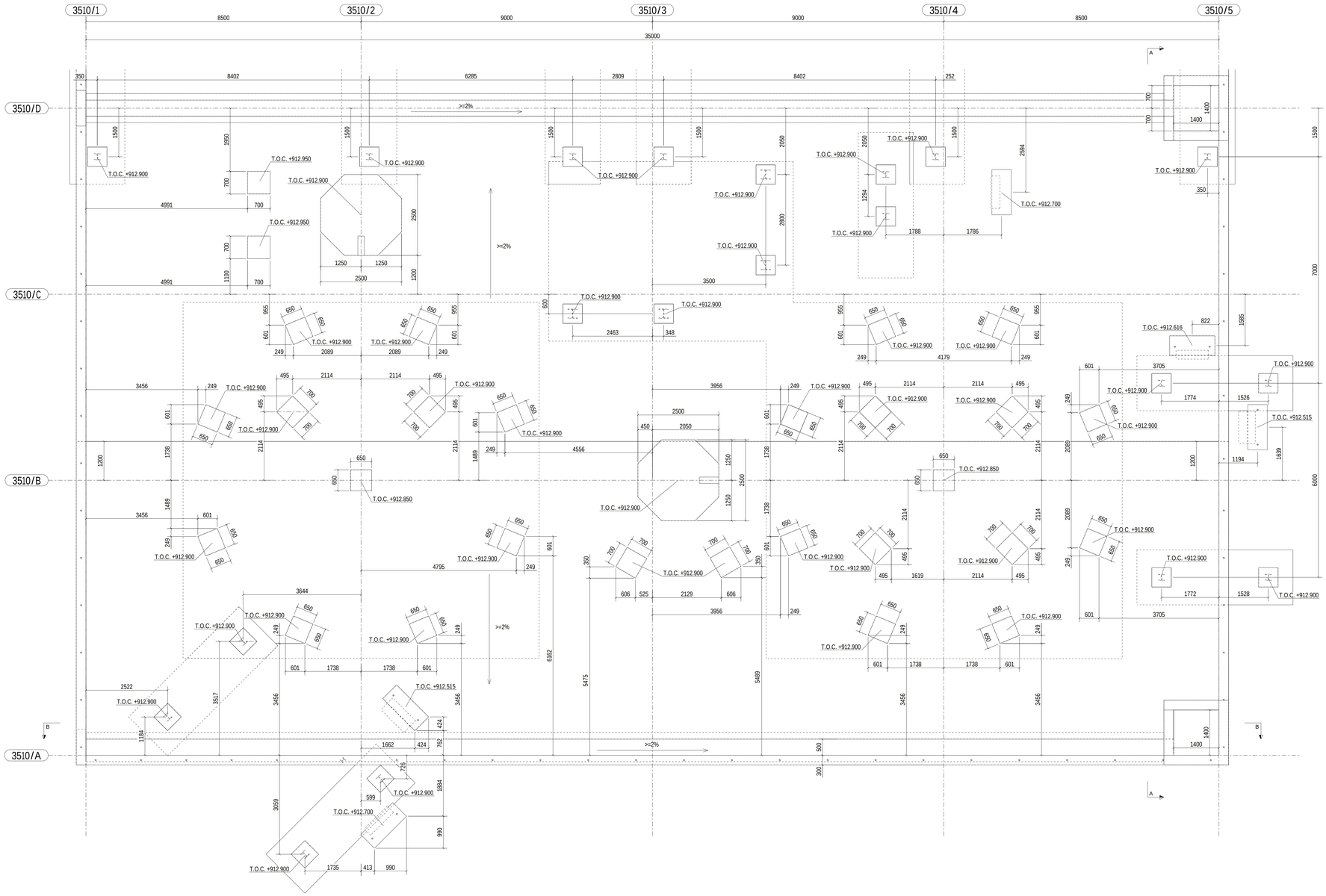
ATTACHMENT 2 OTE INPUT DRAWINGS(CIVIL WORKS GUIDE DRAWING)

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DRAWING NUMBER MD-216-3510-EG-CV-GAD-1001
REV A01

LEVEL +912.500
1:50



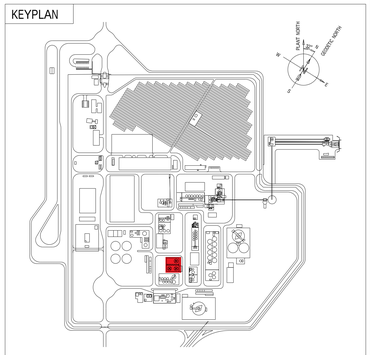
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L)
FOR PROCESS AREA 3510 = RL(+)912.500 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



MD-216-3510-EG-ST-GAD-1001	PLAN AT EL. +912.500	MD-216-3510-EG-ST-GAD-1002	3D VIEW AT EL. +912.500
MD-216-3510-EG-ME-GAD-1019	SECTION B-B	MD-216-3510-EG-CV-GAD-1002	PLAN AT EL. +912.500 - GRID D-G/1-5
MD-216-3510-EG-ME-GAD-1017	PLAN VIEW	MD-216-3510-EG-ME-GAD-1018	SECTION A-A
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3510-EG-ME-GAD-1016	PLAN AT EL. +912.500

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
				A01	11.12.2019	ISSUED FOR APPROVAL	BabHam	MkLai	LuiPz	SuoCha	

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER	O19045_450_C_G_24938_001	REVISION 0
OUTOTEC DRAWING NUMBER	900707_GAU01_ST61_CTA04_35101	DOCUMENT ID OU100667795 / REVISION 0

DRAWN	BabHam	11.12.2019
CHECKED	MkLai	11.12.2019
ENGINEER	LuiPz	11.12.2019
APPROVED	SuoCha	11.12.2019
EMPLOYER		
SCALE	1:50	A0

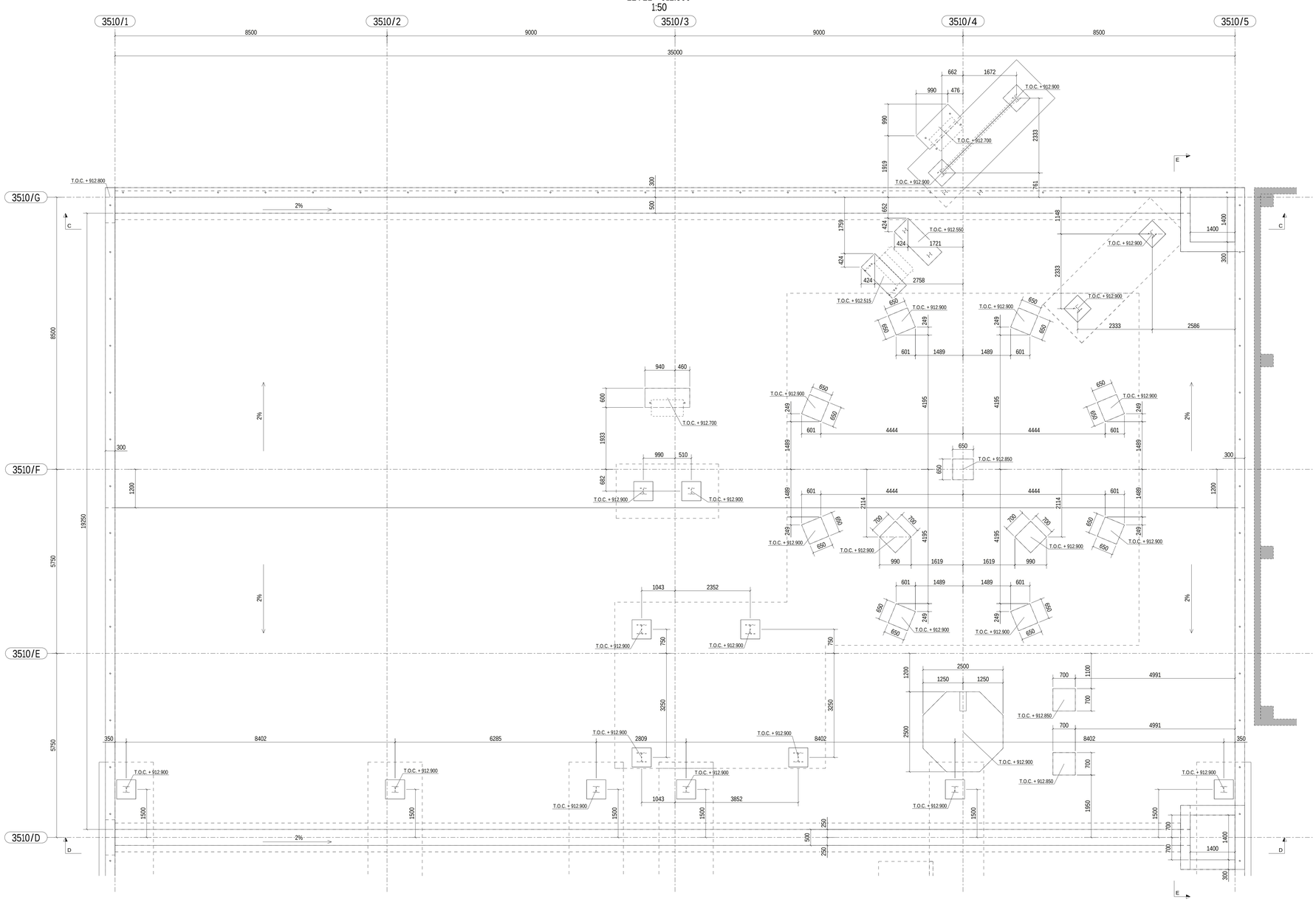
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
CCD CIRCUIT
PLAN AT EL. +912.500 - GRID A-D/1-5

DRAWING NUMBER MD-216-3510-EG-CV-GAD-1001
REV A01

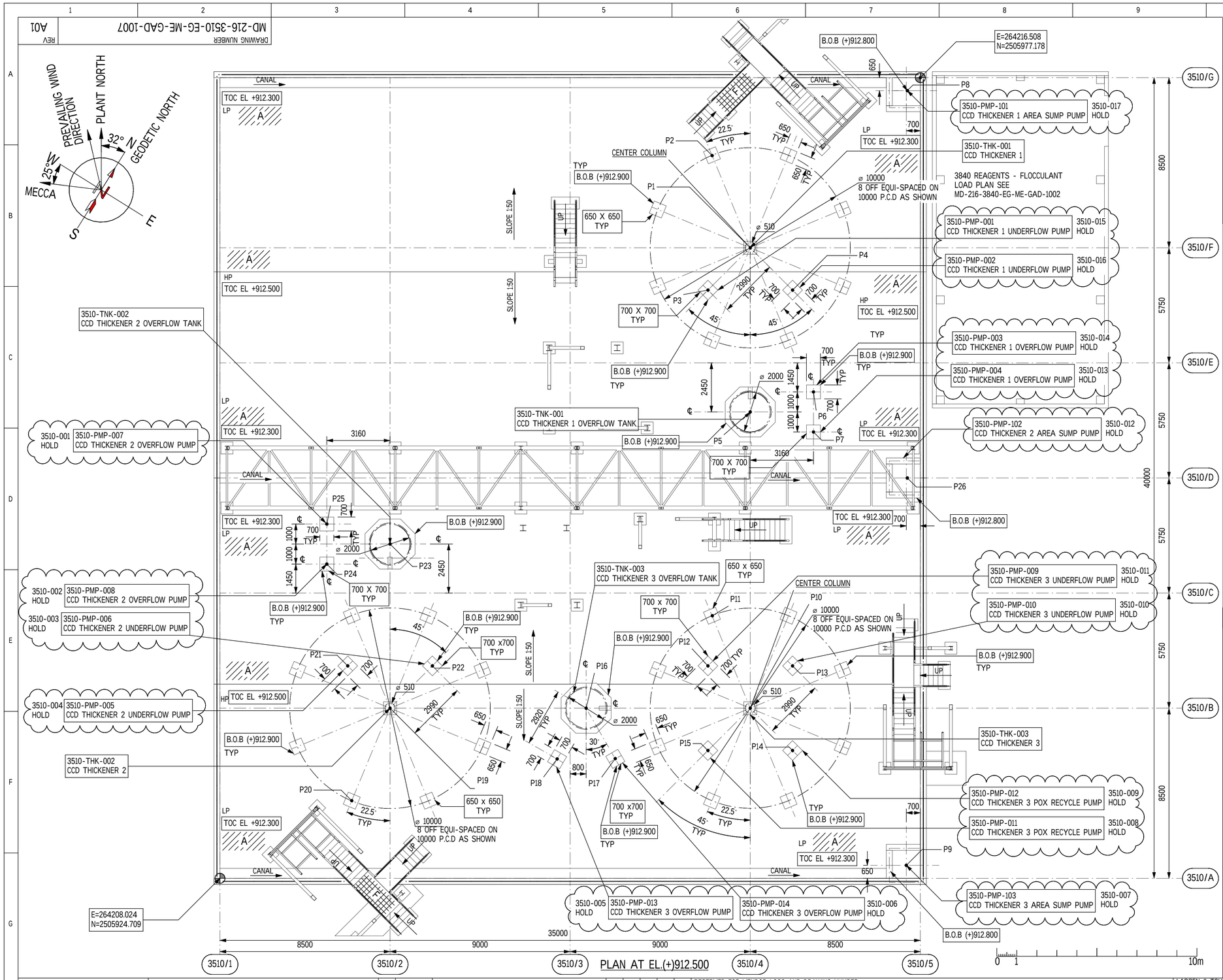
Zaha Structures

DRAWING NUMBER MD-216-3510-EG-CV-GAD-1002 REV A01

LEVEL +912.500
1:50


3510/1
3510/2
3510/3
3510/4
3510/5
3510/G
3510/F
3510/E
3510/D
8500
9000
9000
8500
35000
T.O.C + 912.800
T.O.C + 912.900
T.O.C + 912.700
T.O.C + 912.550
T.O.C + 912.515
T.O.C + 912.850
2%
8500
5750
19250
1200
1500
300
500
250
1400
2333
2586
2758
1721
1672
1619
2333
761
652
1148
1759
1933
682
4195
4444
2114
1489
1619
990
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2352
3250
2500
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1400
300
300
300
300
350
350
250
250
700
700
700

DRAWING NUMBER MD-216-3510-EG-ME-GAD-1007 REV A01



PLAN AT EL.(+)912.500

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
A — CONCRETE FLOOR WITH ACID RESISTANT COATING
CHECKERED PLATE
GRATING
F — GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

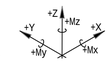
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[KNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES
-WIND LOADS
-EARTHQUAKE LOADS

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

GRATING 5 kN/m² | CONCRETE PLATFORMS 10 kN/m² | GROUNDFLOOR 25 kN/m² | CABLE FLOOR 10 kN/M²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² LIVE LOAD 0.75 kN/m²

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

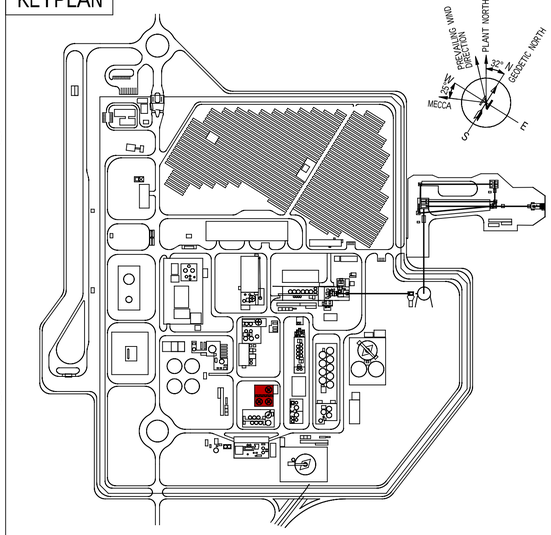
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.500 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3510-EG-ME-GAD-1016	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.500								
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3510-EG-ME-GAD-1009	LOAD TABLE								
MD-216-3510-EG-ME-GAD-1008	PLAN AT EL. +917.000	A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

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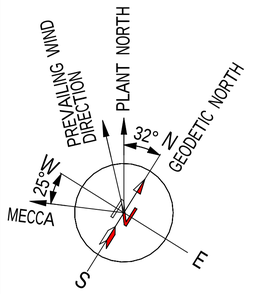
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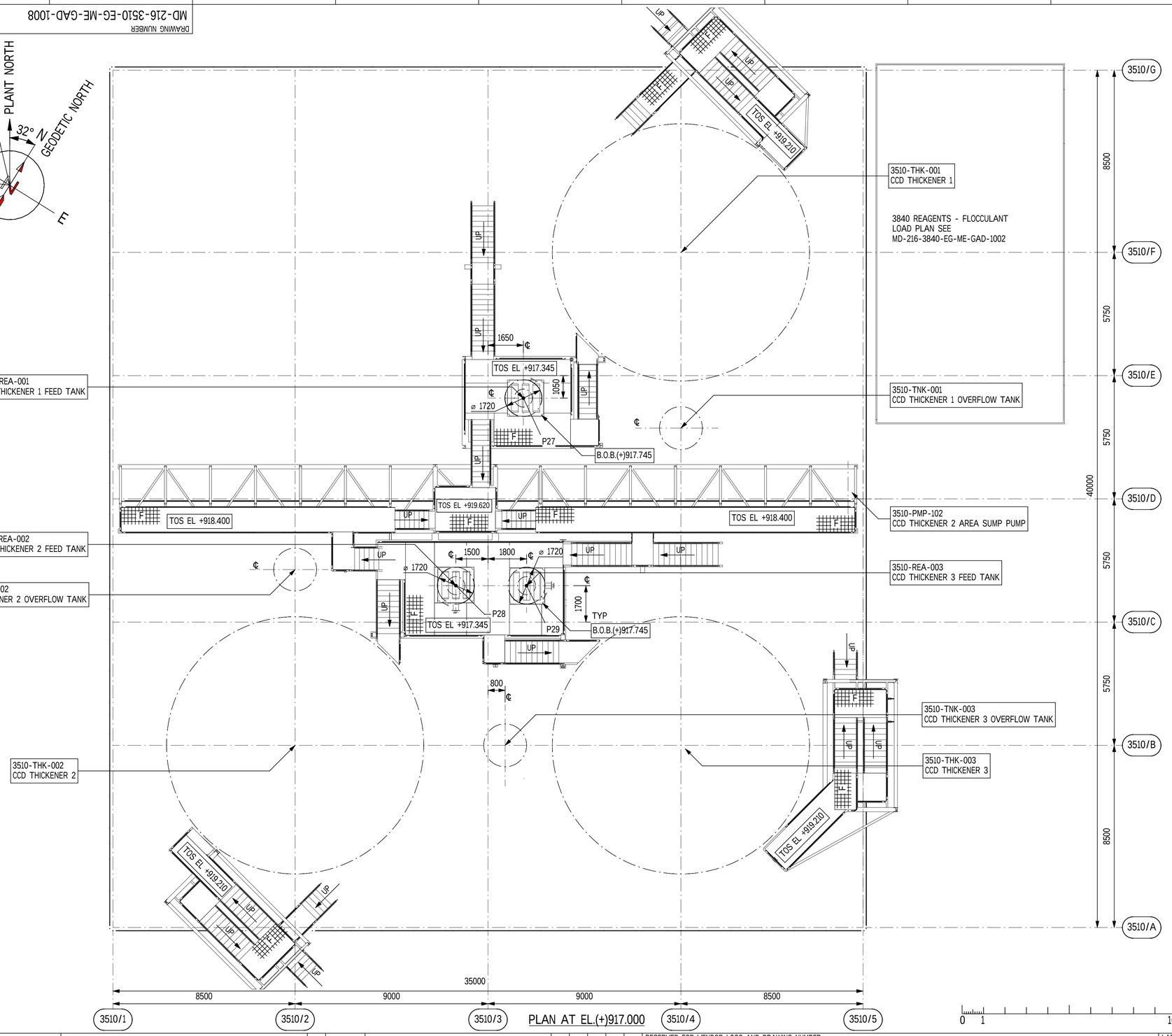
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_M_D_02812_001			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _ST61 _MLD14 _35101		OU100667828	0
DRAWN	BS	03-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03-01-2020	LOAD PLAN
ENGINEER	SC	03-01-2020	CCD CIRCUIT
APPROVED	RL	03-01-2020	PLAN AT EL. +912.500
EMPLOYER			
SCALE 1:100	A1		DRAWING NUMBER MD-216-3510-EG-ME-GAD-1007 REV A01

DRAWING NUMBER MD-216-3510-EG-ME-GAD-1008 REV A01





- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

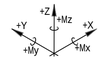
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

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04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

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LP = LOW POINT.
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DIRECTION OF FORCES AND MOMENTS



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DUST 0.5 kN/m² LIVE LOAD 0.75 kN/m²

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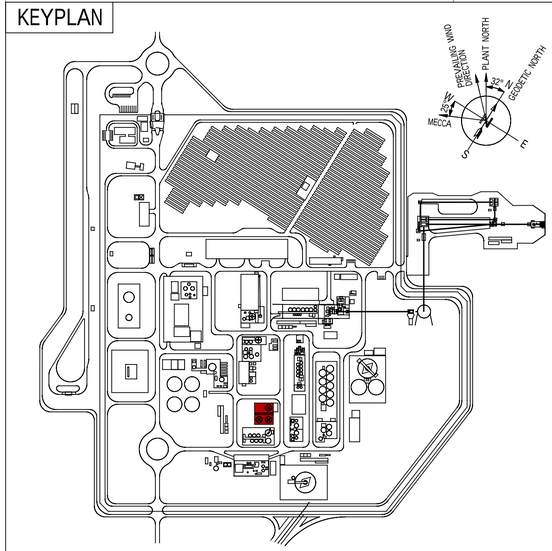
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INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3510-EG-ME-GAD-1016	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.500								
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3510-EG-ME-GAD-1009	LOAD TABLE								
MD-216-3510-EG-ME-GAD-1007	PLAN AT EL. +912.500	A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER | REVISION

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ST61 _MLD14 _35102 | DOCUMENT ID: OU100668574 | REVISION: 0

DRAWN	BS	03-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03-01-2020	LOAD PLAN
ENGINEER	SC	03-01-2020	CCD CIRCUIT
APPROVED	RL	03-01-2020	PLAN AT EL. +917.000
EMPLOYER			
SCALE 1:100		A1	DRAWING NUMBER: MD-216-3510-EG-ME-GAD-1008 REV: A01

DRAWING NUMBER MD-216-3510-EG-ME-GAD-1009 REV A01

* :-
FOR,
P1, P2 : REFER DWG. MD-216-3510-EG-ME-GAD-1002 FOR LOAD DETAILS
P10, P11 : REFER DWG. MD-216-3510-EG-ME-GAD-1006 FOR LOAD DETAILS
P19, P20 : REFER DWG. MD-216-3510-EG-ME-GAD-1004 FOR LOAD DETAILS

	LOADING POINT	P1*				P2*				P3				P4				P5				P6			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT									- 4				- 4				- 12				- 7			
2	DEAD LOAD (SECONDARY EQUIPMENT)																								
3	OPERATION LOAD 1. FILLING																	- 62							
4	OPERATION LOAD 2. PRESSURE																								
5	OPERATION LOAD 3. TEMPERATURE																								
6	OPERATION LOAD 4. DYNAMIC									- 6				- 6								- 11			
7	OPERATION LOAD 5. OTHERS																								
8	TEST LOAD FILLING / EMERGENCY																								
9	LIVE LOAD																								
10	WIND ±X - DIRECTION																								
11	WIND ±Y - DIRECTION																								
12	EARTHQUAKE ±X - DIRECTION																								
13	EARTHQUAKE ±Y - DIRECTION																								
	EQUIPMENT - NO.	3510-THK-001				3510-THK-001				3510-PMP-001				3510-PMP-002				3510-TNK-001				3510-PMP-003			
	DESCRIPTION	CCD THICKENER 1				CCD THICKENER 1				CCD THICKENER 1 UNDERFLOW PUMP				CCD THICKENER 1 UNDERFLOW PUMP				CCD THICKENER 1 OVERFLOW TANK				CCD THICKENER 1 OVERFLOW PUMP			

	LOADING POINT	P7				P8				P9				P10*			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	- 7				- 8				- 8							
2	DEAD LOAD (SECONDARY EQUIPMENT)																
3	OPERATION LOAD 1. FILLING																
4	OPERATION LOAD 2. PRESSURE																
5	OPERATION LOAD 3. TEMPERATURE																
6	OPERATION LOAD 4. DYNAMIC	- 11				- 12				- 12							
7	OPERATION LOAD 5. OTHERS																
8	TEST LOAD FILLING / EMERGENCY																
9	LIVE LOAD																
10	WIND ±X - DIRECTION																
11	WIND ±Y - DIRECTION																
12	EARTHQUAKE ±X - DIRECTION																
13	EARTHQUAKE ±Y - DIRECTION																
	EQUIPMENT - NO.	3510-PMP-004				3510-PMP-101				3510-PMP-103				3510-THK-003			
	DESCRIPTION	CCD THICKENER 1 OVERFLOW PUMP				CCD THICKENER 1 AREA SUMP PUMP				CCD THICKENER 3 AREA SUMP PUMP				CCD THICKENER 3			

	LOADING POINT	P11*				P12				P13				P14				P15				P16			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT					- 4				- 4				- 2				- 2				- 12			
2	DEAD LOAD (SECONDARY EQUIPMENT)																								
3	OPERATION LOAD 1. FILLING																					- 62			
4	OPERATION LOAD 2. PRESSURE																								
5	OPERATION LOAD 3. TEMPERATURE																								
6	OPERATION LOAD 4. DYNAMIC					- 6				- 6				- 3				- 3							
7	OPERATION LOAD 5. OTHERS																								
8	TEST LOAD FILLING / EMERGENCY																								
9	LIVE LOAD																								
10	WIND ±X - DIRECTION																								
11	WIND ±Y - DIRECTION																								
12	EARTHQUAKE ±X - DIRECTION																								
13	EARTHQUAKE ±Y - DIRECTION																								
	EQUIPMENT - NO.	3510-THK-003				3510-PMP-009				3510-PMP-010				3510-PMP-012				3510-PMP-011				3510-TNK-003			
	DESCRIPTION	CCD THICKENER 3				CCD THICKENER 3 UNDERFLOW PUMP				CCD THICKENER 3 UNDERFLOW PUMP				CCD THICKENER 3 POX RECYCLE PUMP				CCD THICKENER 3 POX RECYCLE PUMP				CCD THICKENER 1 OVERFLOW PUMP			

	LOADING POINT	P17				P18				P19*				P20*			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	- 4				- 4											
2	DEAD LOAD (SECONDARY EQUIPMENT)																
3	OPERATION LOAD 1. FILLING																
4	OPERATION LOAD 2. PRESSURE																
5	OPERATION LOAD 3. TEMPERATURE																
6	OPERATION LOAD 4. DYNAMIC	- 6				- 6											
7	OPERATION LOAD 5. OTHERS																
8	TEST LOAD FILLING / EMERGENCY																
9	LIVE LOAD																
10	WIND ±X - DIRECTION																
11	WIND ±Y - DIRECTION																
12	EARTHQUAKE ±X - DIRECTION																
13	EARTHQUAKE ±Y - DIRECTION																
	EQUIPMENT - NO.	3510-PMP-014				3510-PMP-013				3510-THK-002				3510-THK-002			
	DESCRIPTION	CCD THICKENER 3 OVERFLOW PUMP				CCD THICKENER 3 OVERFLOW PUMP				CCD THICKENER 2				CCD THICKENER 2			

	LOADING POINT	P21				P22				P23				P24				P25				P26			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	- 4				- 4				- 12				- 7				- 7				- 8			
2	DEAD LOAD (SECONDARY EQUIPMENT)																								
3	OPERATION LOAD 1. FILLING									- 62															
4	OPERATION LOAD 2. PRESSURE																								
5	OPERATION LOAD 3. TEMPERATURE																								
6	OPERATION LOAD 4. DYNAMIC	- 6				- 6								- 10				- 10				- 12			
7	OPERATION LOAD 5. OTHERS																								
8	TEST LOAD FILLING / EMERGENCY																								
9	LIVE LOAD																								
10	WIND ±X - DIRECTION																								
11	WIND ±Y - DIRECTION																								
12	EARTHQUAKE ±X - DIRECTION																								
13	EARTHQUAKE ±Y - DIRECTION																								
	EQUIPMENT - NO.	3510-PMP-005				3510-PMP-006				3510-TNK-002				3510-PMP-008				3510-PMP-007				3510-PMP-102			
	DESCRIPTION	CCD THICKENER 2 UNDERFLOW PUMP				CCD THICKENER 2 UNDERFLOW PUMP				CCD THICKENER 1 OVERFLOW TANK				CCD THICKENER 2 OVERFLOW PUMP				CCD THICKENER 2 OVERFLOW PUMP				CCD THICKENER 2 AREA SUMP PUMP			

	LOADING POINT	P27				P28				P29			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	- 30				- 30				- 30			
2	DEAD LOAD (SECONDARY EQUIPMENT)												
3	OPERATION LOAD 1. FILLING	- 50				- 50				- 50			
4	OPERATION LOAD 2. PRESSURE												
5	OPERATION LOAD 3. TEMPERATURE												
6	OPERATION LOAD 4. DYNAMIC												
7	OPERATION LOAD 5. OTHERS												
8	TEST LOAD FILLING / EMERGENCY												
9	LIVE LOAD	- 55				- 55				- 55			
10	WIND ±X - DIRECTION												
11	WIND ±Y - DIRECTION												
12	EARTHQUAKE ±X - DIRECTION												
13	EARTHQUAKE ±Y - DIRECTION												
	EQUIPMENT - NO.	3510-REA-001				3510-REA-002				3510-REA-003			
	DESCRIPTION	CCD THICKENER 1 FEED TANK				CCD THICKENER 2 FEED TANK				CCD THICKENER 3 FEED TANK			

0 1 10m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING (A)
CHECKERED PLATE
GRATING
GRATING - FRP (F.R.P.)
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

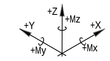
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

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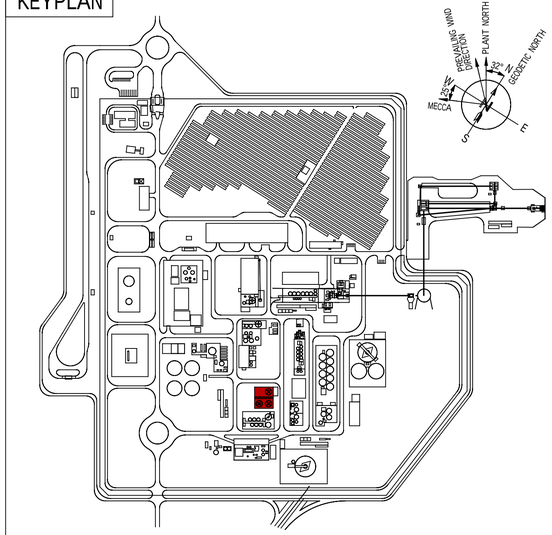
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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3510-EG-ME-GAD-1016	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.500								
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3510-EG-ME-GAD-1008	PLAN AT EL. +917.000								
MD-216-3510-EG-ME-GAD-1007	PLAN AT EL. +912.500	A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_D_02871_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707 _GAU01 _ST61 _MLD14 _35103	DOCUMENT ID OU100669095	REVISION 0
DRAWN	BS	03-01-2020	
CHECKED	TV	03-01-2020	
ENGINEER	SC	03-01-2020	
APPROVED	RL	03-01-2020	
EMPLOYER			
SCALE	NA	A1	

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
CCD CIRCUIT
LOAD TABLE

DRAWING NUMBER MD-216-3510-EG-ME-GAD-1009 REV A01

ATTACHMENT 3 COMMENT RESOLUTION SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-001183, 31 January 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 31 January 2020	Comment by	EPC Response DATE: 30 March 2020	Status	Remarks
			General Comments (cc)				
MD-216-3510-EG-CV-CAL-1280-A01	3510	1	use appropriate SI, kN	Mr. Doron Golan - Bechtel	Included in the revised calculation.	Closed	
MD-216-3510-EG-CV-CAL-1280-A01	3510	2	Shear force on 4 columns: Total seismic force = 6*4 + 10 = 34 different than 37	Mr. Doron Golan - Bechtel	The design of tank foundation has been carried out based on approved equipment foundation load data. However, as per our understanding, there are tatal 8 nos outer columns, out which 4 columns are braced along X-direction and 4 columns are braced along Y-direction. Maximum shear force for each direction are taken by the columns braced along that direction. But the other columns not braced along the direction of force are taking the balance small amount of shear force (approximately 0.75 KN each).	Closed	
MD-216-3510-EG-CV-CAL-1280-A01	3510	3	Seismic coefficient = 37/(144+3825)= 0,0093 Please verify / confirm this value	Mr. Doron Golan - Bechtel	As confirmed by OTE, the seismic forces are checked and ok.	Closed	